VIT

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

CHENNAI

22BAI1066

## Final Assessment Test(FAT) - NOV/DEC 2025

| | | | |
|---|---|---|---|
| Programme | **B.Tech.** | Semester | **Fall Semester 2025-26** |
| Course Code | **BCSE416L** | Faculty Name | **Prof. Graceline Jasmine S** |
| Course Title | **Game Programming** | Slot | **C1+TC1** |
| | | Class Nbr | **CH2025260102071** |
| Time | **3 hours** | Max. Marks | **100** |

**Instructions To Candidates**

- Write only your registration number in the designated box on the question paper. Writing anything elsewhere on the question paper will be considered a violation.

**Course Outcomes**

CO1: Design, develop, test, evaluate, debug, and modify code to meet design specifications for games.

CO2: Design unique gaming environments, levels and characters by choosing appropriate game strategies and patterns based on an analysis of past and present trends.

CO3: Create and document the games by applying programming concepts using various tools to meet requirements of the current marketplace.

**Answer all Questions (10 × 10 Marks)**

01. Action-adventure games combine fast-paced battle, exploration, and puzzle-solving with rich storylines and interactive environments. They rely on dynamic worlds, real-time physics, and meaningful character interactions to engage players. Choose one action-adventure game, and answer the following:

(a) Mention the game premise of the game chosen and list five essential components of the game engine's runtime architecture that are required to handle real-time gameplay interactions, such as character movement, fight, and environment responses. [5 Marks]

(b) For each component, explain its purpose and how it contributes to: [5 Marks]

- Player immersion
- Smooth and responsive interactions
- Real-time behavior of the game world

**[10] (CO2/K2)**

02. Adventure Quest is an open-world role-playing game (RPG) where players interact with multiple characters, collect items, and solve text-based puzzles. During testing, the development team noticed challenges in, in-game text and dialogue that are displayed and managed.

(a) List 5 possible string-related issues that may occur during the development of such a game. [5 Marks]

(b) For each issue listed in part (a), propose a string handling technique or programming solution that can be implemented to ensure accurate, maintainable, and consistent text output. Provide justification for your solution. [5 Marks]

**[10] (CO1/K4)**

03. In a 3D adventure game, the player finds a magical crystal at position (1, 2, 3). When the crystal is activated, the following changes happen:

(a) The crystal is rotated 30° around the Y-axis (using the world origin as the center) and then moved by a translation vector T(–2, 1, 4). Determine the new position of the crustal. [5 Marks]

(b) After the above transformations, the crystal magically grows such that its position becomes 3 times farther from the origin. Compute the final coordinates of the crystal. [5 Marks]

**[10] (CO2/K3)**

04. In a 3D racing game, the player drives a car through a highly detailed city environment with reflective glass buildings, dynamic sunlight, and particle effects like smoke and dust. The game engine uses a rendering pipeline to produce realistic visuals while maintaining smooth performance.

(a) Illustrate the sequence of stages in the rendering pipeline that are executed for each frame in this game. Explain the purpose of each stage and justify how it contributes to visual quality and real-time performance. [5 Marks]

(b) Analyse three key stages in the pipeline that are especially critical for achieving realistic lighting, shadows, and reflections. Justify why each of these stages is essential for maintaining both visual realism and performance in a real-time racing environment. [5Marks]

**[10] (CO1/K3)**

05. In a 3D adventure game, the player discovers a magical shield that reflects light from a nearby torch in a dark dungeon. At a specific point on the shield, the surface has a normal vector N = (0, 3, 4), the light from the torch comes from direction L = (2, 0, 2), and the viewer is positioned such that the viewing direction is V = (0, 0, 5). The shield's material properties are: ambient reflection coefficient k_a = 0.2, diffuse reflection coefficient k_d = 0.6, and specular reflection coefficient k_s = 0.5. The ambient, diffuse, and specular light intensities are all 1, and the shininess factor is α = 2.
Calculate the total light intensity at this point on the shield using the Phong reflection model by performing the following:
(a) Determine the reflection vector R for the surface. [5 Marks]
(b) Calculate the ambient, diffuse, and specular contributions and then compute the total light intensity I. [5 Marks]

**[10] (CO2/K3)**

06. A girl pushes a small toy car across a smooth wooden floor. The toy car has a mass of 200 grams, and the coefficient of friction between the car and the floor is 0.05. If the car is given an initial speed of 5 m/s.
(a) Calculate the distance it will travel before coming to rest. Show all the necessary steps clearly. [5 Marks]
(b) Propose any three testing mechanisms that can be used to verify the functionality and accuracy of the toy car's motion and stopping behavior. Explain briefly how each mechanism ensures reliable testing results.[5 Marks]

**[10] (CO1/K3)**

07. A drone is assigned to deliver a medical package across a large city filled with skyscrapers, bridges, and restricted air zones. Before takeoff, the drone's navigation system loads a complete 3D map of the environment, including all building heights and no-fly zones. The drone must compute the shortest and safest flight path from the dispatch center to the hospital while avoiding obstacles and minimizing travel time.
(a) Which search algorithm would be most suitable for determining this route, and why? Justify. [5 Marks]
(b) In a simulation game where players control delivery drones, the other drones which are AI-controlled act as non-player characters (NPCs) with different roles such as some assist in navigation, while others compete for airspace or deliver packages simultaneously. As an AI designer, how would you design the NPC drones' behavior? Explain your design approach briefly with an example. [5 Marks]

**[10] (CO2/K4)**

08. You are the lead designer for a Virtual Reality (VR) emergency response training program for firefighters. In the simulation, trainees navigate a virtual burning building, locate victims, operate fire suppression equipment, and coordinate with other team members. The environment includes multiple rooms, staircases, obstacles, and dynamic hazards such as smoke, falling debris, and spreading flames.
(a) Design suitable navigation interfaces that allow trainees to move efficiently and safely through the complex, multi-level environment. Analyze how your design ensures realism, spatial awareness, and comfort, avoiding fatigue or motion sickness. [5 Marks]
(b) Formulate appropriate manipulation interfaces that allow trainees to interact intuitively with equipment, doors, and other objects under high-pressure situations. Evaluate how your design provides immediate sensory or instructional feedback and maintains engagement and effectiveness of the simulation. [5 Marks]

**[10] (CO2/K4)**

09. You are developing an AR-based city tourism app that provides users with interactive virtual content as they explore a historical district. The app should display contextual information about nearby landmarks based on the user's location and accurately overlay virtual content on statues and historical signs when viewed through the device.During testing, two challenges are observed. First challenge is where the users sometimes receive information for landmarks they are not physically near and the other is virtual content does not consistently align with statues or historical signs, reducing immersion.
(a) Analyze and evaluate a suitable tracking approach that can effectively address both challenges. Explain how the approach works and how it ensures accurate spatial alignment and relevant content delivery relative to the user's real-world position. [5 Marks]
(b) Justify and assess why the chosen tracking approach is the most appropriate for achieving a realistic and immersive AR experience. [ 5 Marks]

**[10] (CO2/K4)**

10. You are developing a 2D action adventure game in Unity game engine, where the player controls a hero who fights enemies in real time.

(a) Write a C# code snippet to handle player attack include detecting attack input, applying damage to the enemy, and updating enemy health. Include clear comments explaining each step. [7 Marks]
(b) Identify and breif the software engineering method that uses iterative adjustments and implementing changes to refine the player's attack mechanics. [3 Marks]

**[10] (CO1/K3)**

**BL-Bloom's Taxonomy Levels - (K1-Remembering,K2-Understanding,K3-Applying,K4-Analysing,K5-Evaluating,K6-Creating)**